

Playwright Automation Framework – Flight · one-way · cabin

1. Project Title

UI Automation Testing Framework using Playwright – Flight · one-way · cabin

2. Objective

The objective of this project is to design and implement a scalable UI automation framework using Playwright and TypeScript to validate Booking a one way flight.

Target Website:

tripstack.doomple.com

3. User Story

User Story – Booking a flight – one way

As a registered user,
I want to log in using valid credentials,
I want to search for flights,
select a flight and click book now,
fill the passenger details,
fill the card details and pay,
I want to see booking successful page

4. Environment & Setup

Hardware

- **System: Windows**
- **RAM: Minimum 8 GB**
- **Processor: Intel i5 or above**

Software

Component Version

Node.js Latest LTS

Playwright Latest

TypeScript Latest

Component Version

VS Code **Latest**

Browser **Chromium**

5. Framework Design

Framework Architecture:

Fixture Layer

↓

Flow Layer

↓

Page Layer

↓

Test Layer

Description:

Fixtures

Responsible for:

- **Creating reusable test context**
 - **Injecting page objects**
-

Flow Layer

Responsible for:

- **Business workflow implementation**
 - **Combining page actions**
-

Page Layer

Responsible for:

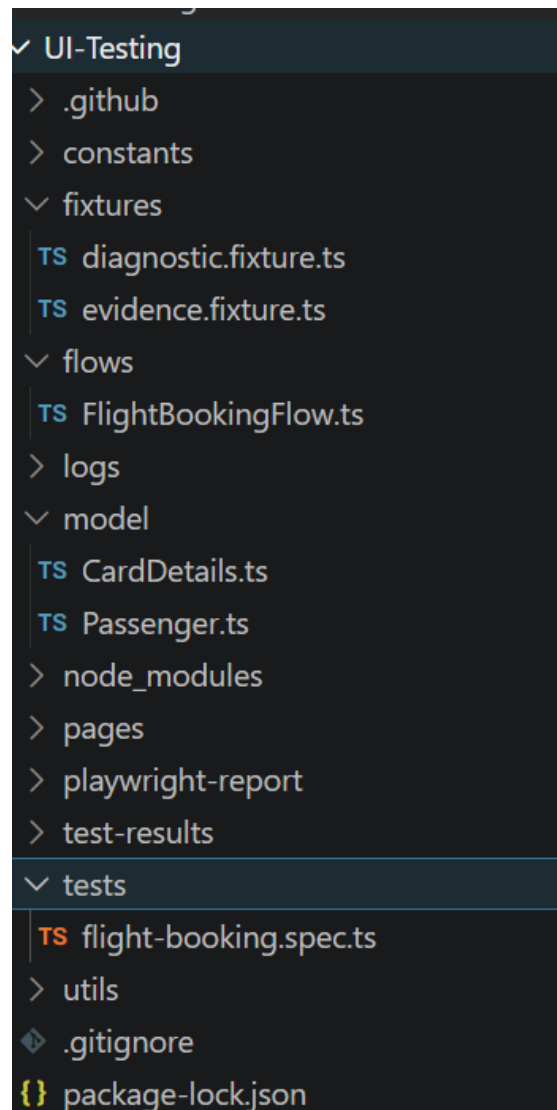
- **Encapsulating locators**
 - **Maintaining UI interactions**
-

Test Layer

Responsible for:

- Test orchestration
 - Assertions
-

6. Project Structure



7. Scenarios Covered

Scenario – Successful Flight Booking

Status:

PASS

8. Execution Steps

Install dependencies:

npm install

Install browsers:

npx playwright install

Execute tests:

npx playwright test website-tests2/tests --headed

Open report:

npx playwright show-report

9. Test Results

Execution Summary:

Total Tests: 1

Passed: 1

Failed: 0

Execution Mode:

Chromium Browser

Execution Duration:

~12 seconds

10. Evidence

Execution Output:

Running 1 tests using 1 workers

```
[INFO] 2026-07-17 15:12:23 - test started
[INFO] 2026-07-17 15:12:23 - Opening Home Page
[INFO] 2026-07-17 15:12:23 - Opening url: /
[INFO] 2026-07-17 15:12:24 - Opening Login Page
[INFO] 2026-07-17 15:12:24 - Clicking element
[INFO] 2026-07-17 15:12:24 - Entering value: sybil@tripstack.test
[INFO] 2026-07-17 15:12:24 - Entering value: Password@123
[INFO] 2026-07-17 15:12:24 - Clicking element
[INFO] 2026-07-17 15:12:25 - Searching for Flight
[INFO] 2026-07-17 15:12:25 - Entering value: CCU
[INFO] 2026-07-17 15:12:25 - Entering value: BOM
[INFO] 2026-07-17 15:12:25 - Entering value: 2026-08-10
[INFO] 2026-07-17 15:12:25 - Clicking element
[INFO] 2026-07-17 15:12:26 - Verifying flight results are displayed
[INFO] 2026-07-17 15:12:26 - Number of flight cards: 8
[INFO] 2026-07-17 15:12:26 - Clicking Book button
[INFO] 2026-07-17 15:12:26 - Clicking element
[WARN] 2026-07-17 15:12:31 - Seat 1A could not be selected. Trying next seat...
[INFO] 2026-07-17 15:12:31 - Selected seat: 1B
[INFO] 2026-07-17 15:12:31 - Clicking element
[INFO] 2026-07-17 15:12:31 - Filling passenger details
[INFO] 2026-07-17 15:12:31 - Entering value: Mani
[INFO] 2026-07-17 15:12:31 - Entering value: Prasad
[INFO] 2026-07-17 15:12:31 - Entering value: 25
[INFO] 2026-07-17 15:12:31 - Entering value: mani@example.com
[INFO] 2026-07-17 15:12:31 - Proceeding to payment
[INFO] 2026-07-17 15:12:31 - Clicking element
[INFO] 2026-07-17 15:12:32 - Entering payment details
[INFO] 2026-07-17 15:12:32 - Entering value: Mani Prasad
[INFO] 2026-07-17 15:12:32 - Entering value: 3647848312340967
[INFO] 2026-07-17 15:12:32 - Entering value: 23/30
[INFO] 2026-07-17 15:12:32 - Entering value: 454
[INFO] 2026-07-17 15:12:32 - Clicking Pay button
[INFO] 2026-07-17 15:12:32 - Clicking element
[INFO] 2026-07-17 15:12:32 - Verifying booking confirmation page
[INFO] 2026-07-17 15:12:32 - test finished {"status":"passed"}
```

Trace:

Enabled on first retry

11. Environment Configuration

Playwright Configuration:

- **Browser: Chromium**
- **Trace Collection: Enabled**
- **Base URL Configured**
- **HTML Reporting Enabled**

Configuration Parameters:

- **fullyParallel**
 - **retries**
 - **workers**
 - **trace**
 - **baseURL**
-

14. Conclusion

The Playwright automation framework was successfully implemented using Page Object Model, Flow Layer, and Fixtures.

The framework achieved:

- **Reusable architecture**
- **Improved maintainability**
- **Reduced code duplication**
- **Automated validation of booking flight functionality**
- **Execution reporting and evidence generation**